Time: 25 Minutes

You will extend your Smart Irrigation implementation to model a sudden pump fault followed by recovery, and analyse the quality and cost of the system response.

Task 1: Error Function
Add a method u_error_B(u) that takes any input signal and returns a corrupted version:
$u_{corrupted}(t) = u(t) + e(t)$

where the error models a temporary pump dropout followed by noisy recovery:

$e(t) = -u(t)$, $4 \le t < 7$ (complete pump failure)
$e(t) = 0.1 \cdot N(0,1)$, $7 \le t < 10$ (noisy restart, use seed=42)
$e(t) = 0$, otherwise

The dropout window zeroes out the pump entirely for 3 seconds.
The noisy recovery models unstable flow during restart.

Apply this to the Step input and Sinusoidal input only.

Task 2: New Metrics
Implement the following two methods:

A. ***Integral of Time-Weighted Absolute Error (ITAE)***

$$ITAE = \int t \cdot |h_{ss} - h(t)|\, dt$$

where h_ss is the steady-state value of the output signal.

B. ***Total Variation (TV)***

$$TV = \Sigma\, |h[n+1] - h[n]|, \quad n = 0, 1, ..., N-2$$

Sum of absolute differences between consecutive samples which measures signal smoothness.

Task 3: Plot
For each of the two inputs (Step, Sinusoidal), produce a figure with two subplots side by side:
Left: Laplace simulation with clean input
Right: Laplace simulation with corrupted input
Both plots should show the input signal and output h(t). Print the **two new metrics** for both the **clean and corrupted responses**.

Helper Functions:
rng = np.random.default_rng(seed=42)
After that, *rng.standard_normal(size)* draws samples from the standard normal distribution.